Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

+91 88008-12254 | garvarora0205@gmail.com | linkedin.com/in/gaminization | github.com/gaminization | garvarora.vercel.app

Education

Vellore Institute of Technology (VIT) — **Vellore, India**
Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | **CGPA: 8.36 / 10.0** — *Aug 2023 – Jul 2027*

Gems International School (CBSE) — **Gurugram, India**
Class XII: 76.4% | Class X: 89.3% — *Nov 2016 – Apr 2023*

Work Experience

LG Soft India (LGSI) – HS/ES India Lab — **Noida, India**
Living Solution Control Development Intern — *Jun 2026 – Jul 2026*
- Wrote C/C++ embedded control logic and state machine handlers for smart appliance control modules, testing execution timing on target microcontrollers; work certified as "Technically Competent" by LGSI engineering leads.

Samsung PRISM — **Remote**
Data Intelligence Agent Developer (Internship) — *Apr 2026 – Oct 2026*
- Multi-agent data intelligence pipeline built with LangGraph, FastAPI, Redis, and Qdrant to automate dataset discovery, PII redaction (spaCy, Presidio), and license compliance across HuggingFace repositories.

Wissen Baum Engineering Solutions LLP — **Pune, India**
Software Automation Intern — *May 2025 – Jul 2025*
- Automated regression testing using Python BDD (Gherkin, pytest, Playwright), parallelizing GitLab CI runs to speed up build pipelines from 10 to 6 minutes (40% speedup) and cut manual QA work by 80%.

SEDS India (SEDS VIT) — **Vellore, India**
R&D Lead & Autonomous Systems Developer — *Apr 2024 – Jul 2026*
- ROS 2 Nav2 and RTAB-Map 3D Visual SLAM stack built on Intel RealSense D455 depth cameras, generating real-time elevation maps that cut manual rover teleoperation time by 70% in competition.
- Accelerated computer vision pipelines using CUDA GPU kernels, YOLOv8 object detection, and V4L2 zero-copy memory buffers, dropping camera latency by 60% across 4 active video feeds.
- Micro-ROS firmware flashed to ESP32 microcontrollers, combining wheel encoder and IMU telemetry with an Extended Kalman Filter (EKF) to reduce odometry drift down to sub-10 cm accuracy.
- Real-time PyQt5 mission control dashboard written for telemetry tracking, motor control, and autonomous state machine execution during live competition runs.

Projects

teleop-cursor | *ROS 2, Python, rclpy, geometry_msgs/Twist* | pypi.org/project/teleop-cursor — **Feb 2026 – Apr 2026**
- Zero-hardware ROS 2 teleoperation node written in Python that maps mouse cursor displacement on screen to geometry_msgs/Twist velocity commands at 10 Hz for Gazebo robotic simulation.

HayaiOS – Preemptive Bare-Metal RTOS Kernel | *C, ARM Assembly, ARM Cortex-M4* — **Jan 2026 – Mar 2026**
- Preemptive bare-metal RTOS kernel built in C and ARM Assembly for Cortex-M4 microcontrollers, featuring a 1 ms tick scheduler, mutex/semaphore primitives, and hardware abstraction layer (HAL).

Captivity CLI | *Rust, Python, Linux D-Bus, systemd, Socket IPC* | pypi.org/project/captivity-cli — **Nov 2025 – Jan 2026**
- WiFi captive portal login daemon built with Rust, Python, systemd, and D-Bus NetworkManager APIs, featuring keyring credential storage and <50ms HTTP login probing.

3D Reconstruction – IMU-Enhanced KinectFusion | *C++, OpenCV, Intel RealSense SDK* — **Aug 2025 – Oct 2025**
- 3D reconstruction pipeline fusing RealSense D455 depth data with 6-DOF IMU telemetry using complementary filtering, reducing tracking failures by 50% during fast camera motion.

Patents

Patent Published (App IN202641072249 A1): *Autonomous Radar Survivor Detection* — **Jun 2026**
- Autonomous radar survivor detection system fusing 24GHz FMCW mmWave radar, ultrasonic sensors, and IMU dead-reckoning on dual-core ESP32; created confidence grid decay formula $C = \max(0, C - \beta\Delta t)$ for survivor breathing detection under rubble.

Achievements & Awards

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (∼2.3% acceptance rate).

International Rover Challenge (IRC): Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT).

Certifications

AWS Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner

Oracle Cloud Certifications: Certified Data Science Professional | Certified Generative AI Professional

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS